

Orthogonally quantifying chemical contamination; Validating least-contaminated (reference) sites; Identifying and extending natural community clusters from reference sites to all sites.

Summaries of Chapter 2 & 3

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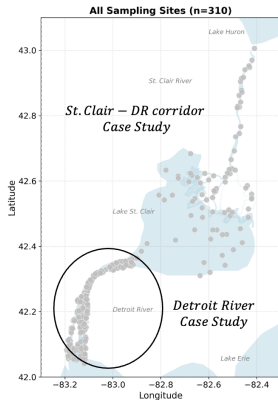
Progress Update

Overview of the Whole Thesis

Core target: study taxa community distortion extent in terms of contamination levels.

- **Chapter 1** — Quantify **contamination levels** of sites; Validate least-contaminated sites where contamination has minimal impact on community composition.
- **Chapter 2** — Identify natural **taxa clusters** in least-contaminated sites; Assign the natural clusters to all sites by environmental descriptors, equivalently partitioning natural variation in contaminated sites.
- **Chapter 3** — In each community type, measure the **ecological dissimilarity** among sites; scale the dissimilarity among sites into an coordinate system where euclidean distance mostly preserves that dissimilarity.
- **Chapter 4** — In each community type, apply regression analysis (piecewise quantile regression) between the scaled community dissimilarity and contamination levels; detect **ecological threshold(s)** and track back to finer-level taxa and latent stressors.

Study Area and Data Source



Data Source	Symbol	Shape	Unit
Environmental	<i>E</i>	(310 × 6)	array (6 × 1)
Chemical	<i>M</i>	(310 × 16)	array (16 × 1)
Taxa	<i>T</i>	(310 × 16)	<i>octave transformed</i>

- Columns/Features in ***E***: Temperature, Water Depth
- Features in ***M***: Al, As, OCS ...
- Features in ***T*** : Dressina, Nematoda ...

- There are two case studies in this thesis: (1) St. Clair - Detroit River corridor case study; (2) Detroit River case study.
- The same methodology is applied to both case studies, DR case study uses the subset of the data in the corridor case and has one additional environmental feature — estimated *water velocity*.

Chapter 1 — Validating the Selection of Least-Contaminated Reference Sites

Chapter 1 at a Glance: Background, Method, Result

Background

Quantifying contamination level is essential for regressing their effects on taxa and selecting least-contaminated sites to find *natural* community types as benchmarks.

Method

Reduce 16 correlated chemicals to orthogonal latent stressors (PCA), sum them into an integrated score (SumRel), then *test* the least-contaminated set cut-off number with two complementary resampling tests.

Result

The lowest-20% set is significantly cleaner than random sets and carries no internal contamination constraint; a size scan shows the corridor tolerates 20% while the Detroit River does not.

Why “Least-Polluted” Is the Only Defensible Baseline

Natural vs. least-polluted

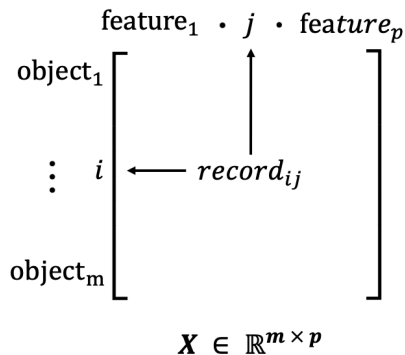
A **natural site** is a theoretical reference state shaped without human stressors — not directly observable here. A **least-polluted site** is operational: defined relative to the observed distribution of integrated chemical stress, taking its low tail as the baseline.

- Within least-polluted sites, community structure is governed by habitat, *not* chemical stress — they retain naturally organised community information.
- They are atoms to constitute the comparable, observable ecological benchmarks for the rest of the thesis.

From 16 Correlated Chemicals to Orthogonal Latent Stressors

- The 16 chemicals are ecologically distinct but **statistically redundant**; correlated variables would carry overlapping information and double-count.
- PCA (R-mode ordination, Euclidean distance) extracts **latent stressors**: mutually orthogonal axes preserving the original stress information.
- Orthogonality guarantees each stressor's signal is uncorrelated with the others, so contamination variation is attributed to the right domain.

Matrix Data Structure: record $\mathbf{X}_{ij}$ stores object i 's value of feature j via pointing paths



How we view a matrix structure in this thesis

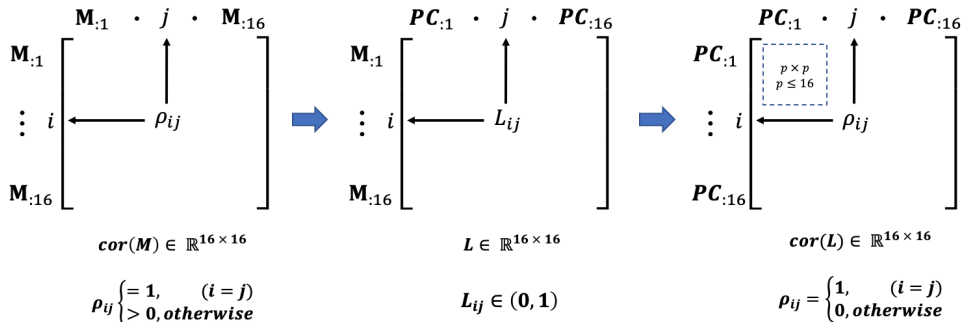
$\mathbf{X} \in \mathbb{R}^{m \times p}$ stores m objects' records of p features:

- Row dimension has m ticks: object i is the i -th tick.
- Column dimension has p ticks: feature j is the j -th tick.
- Entry $\mathbf{X}_{ij}$: a record pointing to object i and feature j .
- $\mathbf{X}_{\cdot j}$: the vector of m records pointing to all m objects for feature j .

This *record-pointing* view is used throughout the thesis and later extends to higher-dimensional tensor structures¹.

¹A matrix is a 2-dimensional tensor: two dimensions store the ticks a record points to, so $\mathbf{X}_{ij} = \mathbf{X}[i, j]$.

Source of overlapping information in **M** and its solution



- In the $\mathbf{M}$ matrix, features $\mathbf{M}_{:i}$ and $\mathbf{M}_{:j}$ are correlated; so the recorded information is double-counted, shown by: $\text{Var}(\sum_{i=1}^{16} \mathbf{M}_{:i}) = \sum_{i=1}^{16} \text{Var}(\mathbf{M}_{:i}) + 2 \sum_{i < j} \text{Cov}(\mathbf{M}_{:i}, \mathbf{M}_{:j})$ ².
- In the resulted loading matrix $\mathbf{L}$ and its PC score matrix $\mathbf{S} = \mathbf{ML}$, $\mathbf{L}_{:i}$ and $\mathbf{L}_{:j}$ are orthogonal, $\text{Var}(\sum_{i=1}^p \mathbf{S}_{:i}) = \sum_{i=1}^p \text{Var}(\mathbf{S}_{:i})$.

²This overlapping-information structure is specifically studied in the field of time series analysis, where a stochastic process $\{X_t\}$ has autocorrelated terms.

Computation steps from chemical matrix **M** to PC Loading matrix-**L** and PC scores matrix-**S**

Algorithm 1: Construction of the PC-based chemical stressor matrix

Data: Site-by-chemical matrix $\mathbf{M}_{310 \times 16}$.

Result: PC-based chemical stressor matrix $\mathbf{S}_{310 \times p}$.

1 Log-transform and standardise, then form the correlation matrix:

2 $\mathbf{M}_{310 \times 16} \rightarrow \mathbf{M}'_{310 \times 16} \xrightarrow{\text{cor}} \text{cor}(\mathbf{M}')_{16 \times 16}$

3 Apply PCA, quartimax rotation (flip afterwards if needed), and loading selection:

4 $\text{cor}(\mathbf{M}')_{16 \times 16} \xrightarrow{\text{pca}} \text{Loadings}_{16 \times p'} \xrightarrow{\text{rotation}} \text{Loadings}^*_{16 \times p'} \xrightarrow{\text{check \& selection}} \text{Loadings}^*_{16 \times p}$

5 Project the processed chemical matrix to the selected loading space:

6 $\mathbf{S}_{310 \times p} = \mathbf{M}'_{310 \times 16} \text{Loadings}^*_{16 \times p}$

7 return $\mathbf{S}_{310 \times p}$

- Log-transformation compresses the extreme values so that they do not improporportionally distort the correlation coefficients $\text{cor}(\mathbf{M})_{ij}, (i \neq j)$ between chemical variables.
- Loading rotation concentrates each PC's loadings on a few chemical variables, so PCs can be interpreted more directly in terms of those dominant chemicals.

Resulting PC Loading matrix L^* (Corridor case)

Chemical	PC ₁	PC ₂	PC ₃	PC ₄	PC ₅	PC ₆
Cr	0.9513	0.1267	0.1178	0.0032	-0.0226	0.0222
Cu	0.8999	0.0993	0.0920	-0.0732	0.0184	0.0117
Ni	0.8819	-0.0435	-0.0654	0.0386	-0.2756	0.0631
Zn	0.8707	-0.1695	-0.0187	-0.0708	0.1240	-0.0418
Fe	0.7923	0.1698	0.0292	0.0245	0.0519	0.2347
SumPCBs	0.7756	-0.4605	0.0212	0.1543	0.0438	-0.0751
Co	0.7635	0.4566	0.0062	0.0725	0.3474	0.0609
Cd	0.7358	-0.2001	0.4249	0.0515	0.1110	0.0586
p,p'-DDE	0.5937	-0.3834	-0.0932	0.3676	0.2515	-0.1579
Mn	0.5897	0.4149	0.3051	0.1630	-0.2672	0.3835
Al	0.5676	0.6718	0.0079	0.0345	0.3298	0.1848
As	0.3392	-0.0176	0.8093	-0.0294	0.2989	0.0364
Pb	0.3933	0.0837	0.7681	0.0369	-0.3819	-0.0015
OCS	0.1905	0.0145	0.0149	0.9444	0.0122	0.1100
Hg	0.4137	0.1400	0.0236	0.0359	0.8039	0.0486
Ca	0.2978	0.0942	0.0199	0.0965	0.0466	0.9299
Explained Var.	7.1925	1.3524	1.5568	1.1119	1.3571	1.1609
Proportion of Var.	0.4495	0.0845	0.0973	0.0695	0.0848	0.0726
Cumulative Var.	0.4495	0.5341	0.6313	0.7008	0.7857	0.8582

Stressor Matrix $\mathbf{S}$ and Collapsing Stressors into One Integrated Score

Of the six rotated PCs, four, loading high on **toxic** chemicals, are retained in $\mathbf{L}^*$. PC_2 and PC_5 are dropped: they load positively on the geogenic elements Al, Ca, and Fe [1], which reflect **natural** lithogenic background rather than contamination [2], and negatively on some toxic chemicals (p,p'-DDE, SumPCBs). Projecting the processed chemical matrix $\mathbf{M}'$ gives four latent stressors per site:

$$\mathbf{S}_{310 \times 4} = \mathbf{M}'_{310 \times 16} \mathbf{L}_{16 \times 4}^*$$

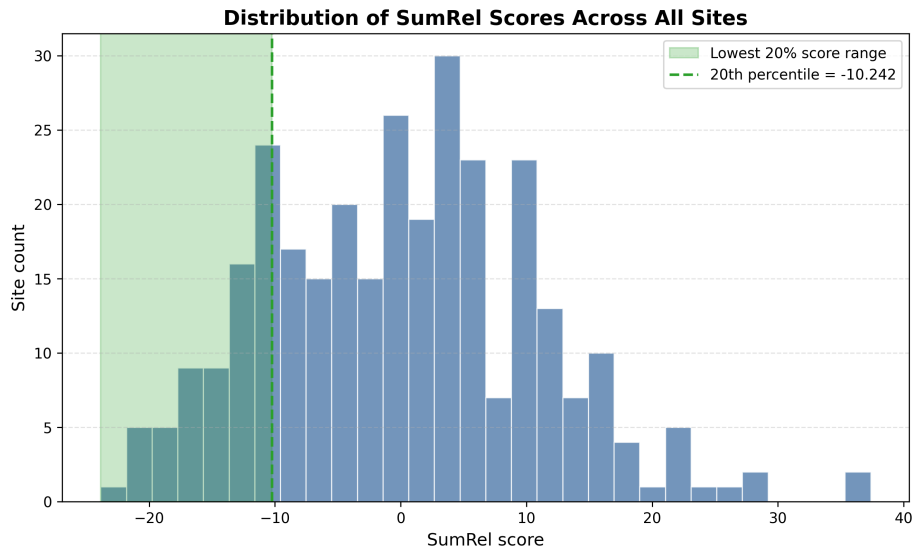
which are summed into one integrated score:

$$\text{SumRel}_i = \sum_{j=1}^4 s_{ij}, \quad s_{ij} \in \mathbf{S}_{310 \times 4}$$

Integrated contamination level and the additive-impact perspective

Summing the latent stressors into SumRel collapses them into a single measure of *integrated* contamination level. Studying the community response against this sum *implicitly assumes* an *additive impact* [3]. Alternative impact views [4] exist, but this study adopts the additive perspective.

Resulting SumRel Histogram and a candidate least-contaminated set



Bridge the Gap between Least-Contaminated and Natural Benchmark

Background and motivation

A 20% lowest cut-off is conventionally used to select least-contaminated sites, they are assumed minimally impacted and set as natural benchmarks. Various contaminated areas have their lowest 20% sites impacted differently. It needs to test whether a cut-off value chooses sites of defensible natural benchmarks and how to adjust it if it did not.

Objectives

- Given a cut-off m_{ref} , answer whether the lowest-scored m_{ref} sites have their taxa community significantly less impacted than a random m_{ref} -unit subset.
- Given a m_{ref} -unit set, answer whether its community is significantly affected by stressor levels. And how likely a random m_{ref} -unit set is affected by stressor levels.
- Answer how the cut-off m_{ref} and the population-level contamination effect simultaneously deciding if a least-contaminated set is defensible as a natural benchmark.

Data, Methods and Theoretical Support

- Taxa matrix **T**, Stressor matrix **S**
- Multivariate Regression; adjusted R^2 ; Resampling tests; Permutation tests.
- Central Limit Theorem; Monte Carlo Simulation; DKW inequality.

Between-group hypothesis test: Are the least-contaminated m_{ref} communities less impacted than random m_{ref} communities?

Hypotheses for the between-group impactation test

H_i	Statement
H_0	The cut-off m_{ref} gives a least-contaminated set η_{ref} that does not explain less community variation than a random m_{ref} -unit subset η_i .
H_A	The cut-off m_{ref} gives a least-contaminated set η_{ref} that explains less community variation than a random m_{ref} -unit subset η_i .

Steps of the test

- Set a cut-off m_{ref} and fetch least-contaminated set η_{ref} . Fit multivariate regression between their taxa composition and latent stressors, $\hat{\mathbf{T}}_{\eta_{\text{ref}}} = \mathbf{S}_{\eta_{\text{ref}}} \beta$. The stressor-explained community variation fraction is $\bar{R}_{\text{ref}}^2 = \frac{SS_{\text{fit}}}{SS_{\text{tot}}}$ after dimensionality adjustment.
- Randomly draw m_{ref} units from the $N = 310$ sites, repeat B times to form a set of random subsets $\{\eta_i\}_{i=1}^B$. Fit the same regression $\hat{\mathbf{T}}_{\eta_i} = \mathbf{S}_{\eta_i} \beta$ for all i and collect the explained variation fractions into $\{\bar{R}_i^2\}_{i=1}^B$.
- Compare $\bar{R}_{\text{ref}}^2$ with the **null distribution** $\{\bar{R}_i^2\}_{i=1}^B$, if $\bar{R}_{\text{ref}}^2$ lies significantly below $\{\bar{R}_i^2\}$, the reference set η_{ref} is significantly less impacted than a random m_{ref} -size subset.

Within-group (permutation) hypothesis test: Is a fixed set of m_{ref} communities internally affected by contamination?

Hypotheses for the within-group impactation test

H_i	Statement
H_0	The m_{ref} sites have their community variation not internally affected by stressor levels.
H_A	The m_{ref} sites have their community variation internally affected by stressor levels.

Steps of the test

- To one m_{ref} -unit set η_i , fix $\hat{\mathbf{T}}_{\eta_i}$ rows, permute $\mathbf{S}_{\eta_i}$ rows with B' times to generate permuted pairs $\{\hat{\mathbf{T}}_{\eta_i}, \mathbf{S}_{\eta_i}^{\pi}\}_{\pi=1}^{B'}$. Fit multivariate regression $\hat{\mathbf{T}}_{\eta_i} = \mathbf{S}_{\eta_i}^{\pi} \beta$ for all permuted B' pairs and collect the explained variation fractions into $\{\bar{R}_{\pi}^2\}_{\pi=1}^{B'}$.
- Compare η_i 's observed $\bar{R}_{\eta_i}^2$ with its **null distribution** $\{\bar{R}_{\pi}^2\}_{\pi=1}^{B'}$, it fails to reject H_0 if $\bar{R}_{\eta_i}^2$ is not significantly higher than $\bar{R}_{\pi}^2$ with a right-tailed p -value $< \alpha$.
- There are B random m_{ref} -unit sets $\{\eta_i\}_{i=1}^B$. Each set η_i is tested with B' permutations, and produces a $\{\bar{R}_{\pi, \eta_i}^2\}_{\pi=1}^{B'}$ with its p_{η_i} -value. The $\{p_{\eta_i}\}_{i=1}^B$ shows how likely a random m_{ref} -unit set fails to reject H_0 or not.

$\hat{\bar{R}}_m^2$ & $\hat{\bar{R}}_\pi^2$: Asymptotic Normality by CLT and DKW Estimation Accuracy

Between-group test. $\bar{R}^2$ is a smooth function of **linear** cross-product quantities computed over the N -unit population¹. By the finite-population CLT [5] and the Delta method [6], the estimate $\hat{\bar{R}}_m^2$ from a random m -unit subset is **asymptotically normal** around the population-level value $\bar{R}^2$:

$$\sqrt{m}(\hat{\bar{R}}_m^2 - \bar{R}^2) \xrightarrow{d} \mathcal{N}(0, \tau^2)$$

For a fixed m , drawing B random m -unit subsets out of the $\binom{N}{m}$ possible sets produces an empirical CDF $\hat{F}_{m,B}(t)$ **approximating** the true subset-level CDF $F_m(t)$. This Monte Carlo approximation [7] has error bounded by the DKW inequality [8, 9]:

$$P\left(\sup_t |\hat{F}_{m,B}(t) - F_m(t)| > \varepsilon\right) \leq 2e^{-2B\varepsilon^2}$$

Within-group test. Fixing one m -size set, there are $m!$ row-permutations. By Hoeffding's combinatorial CLT [10], a single permutation estimate R_π , standardized by the exact closed-form permutation moments μ_S and σ_S of this one set, is **asymptotically standard normal**:

$$\frac{R_\pi - \mu_S}{\sigma_S} \xrightarrow{d} \mathcal{N}(0, 1)$$

where μ_S and σ_S are the mean and standard deviation of R_π taken over all $m!$ permutations of this set. Sampling B' permutations similarly gives an empirical permutation-null $\hat{G}_{m,B'}(t)$ approximating the exact permutation-null $G_S(t)$.

¹ $N = 310$ (corridor) or 213 (DR) is fixed in this study; the statement is used as a finite-sample approximation, justified by m and $N - m$ both being reasonably large, rather than a literal double limit.

Testing Results: Determined by Both Cut-off m_{ref} and Population-Level $\bar{R}^2$

Between-group test. As the cut-off m_{ref} **increases**, the null distribution $\hat{\bar{R}}_m^2$ concentrates more tightly around $\bar{R}^2$ (variance $\sim \tau^2/m$ from the Delta method below), so a given gap between $\bar{R}_{\text{ref}}^2$ and the null mean becomes more standard deviations wide, making H_0 **easier to reject**:

$$\sqrt{m}(\hat{\bar{R}}_m^2 - \bar{R}^2) \xrightarrow{d} \mathcal{N}(0, \tau^2)$$

Conversely, as m_{ref} **decreases**, the null distribution spreads out, and a larger gap is needed to reach the same significance level, making H_0 **harder to reject**. (This holds for a fixed observed gap; in practice $\bar{R}_{\text{ref}}^2$ itself also shifts with m_{ref} , since the reference set's composition changes.)

Within-group test. For one fixed set (size m_{ref} , observed value $\bar{R}_{\text{ref}}^2$), the permutation test gives, for π a random permutation among the $m_{\text{ref}}!$ total:

$$\frac{R_\pi - \mu_S}{\sigma_S} \xrightarrow{d} \mathcal{N}(0, 1)$$

Since the stressor variables are centered, $\mu_S \approx 0$ regardless of m_{ref} and only σ_S depends on m_{ref} . As m_{ref} **increases**, σ_S decreases and a tighter permutation null makes it **harder to not reject H_0** . As m_{ref} **decreases**, σ_S increases, and a wider permutation null makes it **easier to not reject H_0** .

Takeaway

Whether the least-contaminated m_{ref} set passes both tests depends jointly on the cut-off m_{ref} and the population-level $\bar{R}^2$: a less contaminated study area and/or a smaller m_{ref} make both tests easier to pass; a more heavily contaminated area and/or a larger m_{ref} make them harder.

Putting It Together — Scanning the Threshold

Both tests are run at every candidate size, sweeping the least-contaminated set from 15 to 310 sites. The size at which p_{ref} falls below α reveals where contamination begins to constrain the community.

Algorithm 2: Threshold scan across least-contaminated subset sizes

Data: Stressor $\mathbf{X}$, community $\mathbf{Y}$, SumRel-ordered sites, step $\delta = 5$.

Result: Curves $\bar{R}_{\text{ref}}^2(m_{\text{ref}})$ and $p_{\text{ref}}(m_{\text{ref}})$ with bands.

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1 for  $k \leftarrow 0$  to 59 do
2    $m_{\text{ref}} \leftarrow 15 + k \delta$ 
3   Take the  $m_{\text{ref}}$  lowest-SumRel sites; record  $\bar{R}_{\text{ref}}^2$  and  $p_{\text{ref}}$  with random-subset bands
   (Algorithms 2–3)
4 return  $\{\bar{R}_{\text{ref}}^2(m_{\text{ref}})\}$  and  $\{p_{\text{ref}}(m_{\text{ref}})\}$ 
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Appendix: Theoretical Details

Between-group test. Let X_m stack the cross-product matrices (A, B, C) from an m_{ref} -site subset, and θ their population-level (N -site) counterpart. For $m_{\text{ref}}, N - m_{\text{ref}}$ both reasonably large:

$$\sqrt{m_{\text{ref}}}(X_{m_{\text{ref}}} - \theta) \xrightarrow{d} \mathcal{N}(0, \Sigma).$$

Since $\bar{R}^2 = g(X_{m_{\text{ref}}})$ is a smooth function, the Delta method gives

$$\sqrt{m_{\text{ref}}}(\bar{R}^2 - g(\theta)) \xrightarrow{d} \mathcal{N}(0, \nabla g(\theta)^\top \Sigma \nabla g(\theta)).$$

The empirical null from B draws, $\hat{F}_{m_{\text{ref}}, B}$, approximates the true null $F_{m_{\text{ref}}}$ with error bound (DKW):

$$P\left(\sup_t |\hat{F}_{m_{\text{ref}}, B}(t) - F_{m_{\text{ref}}}(t)| > \varepsilon\right) \leq 2e^{-2B\varepsilon^2}.$$

Within-group test. Fixing one m_{ref} -site set, only the cross-term $B^\pi = \mathbf{S}_\eta^\top \mathbf{T}_\eta^\pi$ varies under permutation π ; A, C stay exact constants. By Hoeffding's (1951) combinatorial CLT, with exact closed-form $\mu_\pi = \mathbb{E}[T_\pi]$, $\sigma_\pi^2 = \text{Var}(T_\pi)$:

$$\frac{T_\pi - \mu_\pi}{\sigma_\pi} \xrightarrow{d} \mathcal{N}(0, 1) \quad (m_{\text{ref}} \rightarrow \infty).$$

The empirical permutation null from B' draws, $\hat{G}_{m_{\text{ref}}, B'}$, approximates the exact (fully enumerable at $m_{\text{ref}}!$) permutation-null G_η at the same DKW rate, $O(B'^{-1/2})$.

References: Hájek (finite-population CLT); Hoeffding (1951, combinatorial CLT); Dvoretzky–Kiefer–Wolfowitz (1956, uniform convergence rate).